

A.Introduction

In this project, we develop a machine learning model to predict fish weight based on physical measurements. Unlike digit recognition, which classifies images into discrete categories, this is a **regression problem** where the goal is to estimate a continuous numerical value (weight in grams).

Objective

The goal of this project is to predict the weight of fish based on their physical measurements (length, height, width) and species using machine learning. Accurate weight estimation is crucial in fisheries, aquaculture, and market pricing, where direct weighing may be impractical.

Importance

- **Fisheries Management:** Helps in stock assessment without harming fish.
- **Aquaculture Optimization:** Aids in feeding and growth monitoring and enables real-time growth monitoring without stressing fish.
- **Market Pricing:** Ensures fair trade by estimating weight when scales are unavailable.
- **Research Applications:** Useful in biological studies where repeated weighing is invasive.
- **Resource Efficiency:** Reduces labor costs in large-scale operations.
- **Economic Impact:** Helps farmers optimize feeding schedules and harvest timing

Problem Type

- This is a **supervised regression task**, where we predict a continuous target (**Weight**) from features like length, height, width, and species.

B. Data Exploration (EDA)

Dataset Description

The dataset contains **159 fish samples** across **7 species** with the following features:

- **Target Variable:** Weight (grams)
- **Numerical Features:** Length 1(vertical length in cm), Length 2(diagonal length in cm), Length 3(cross length in cm), Height (cm), Width (cm)
- **Categorical Feature:** Species (7 categories: Bream, Roach, Whitefish, Parkki, Perch, Pike, Smelt)

Key Visualizations

1. Distribution of Weight

- **Right-skewed**, indicating some heavier outliers.
- Most fish weigh between **0–1000g**, with a few exceeding 1500g.

2. Pairplot of Numerical Features

- Strong linear correlations between **Length1**, **Length2**, **Length3**.
- **Height** and **Width** show a moderate relationship with **Weight**.

3. Correlation Matrix

- **Weight** has the highest correlation with:
 - **Width (0.89)**
 - **Height (0.84)**
 - **Length3 (0.83)**

4. Boxplot of Weight by Species

- **Pike and Bream** are the heaviest species.

- **Smelt** are the lightest.

C. Methodology

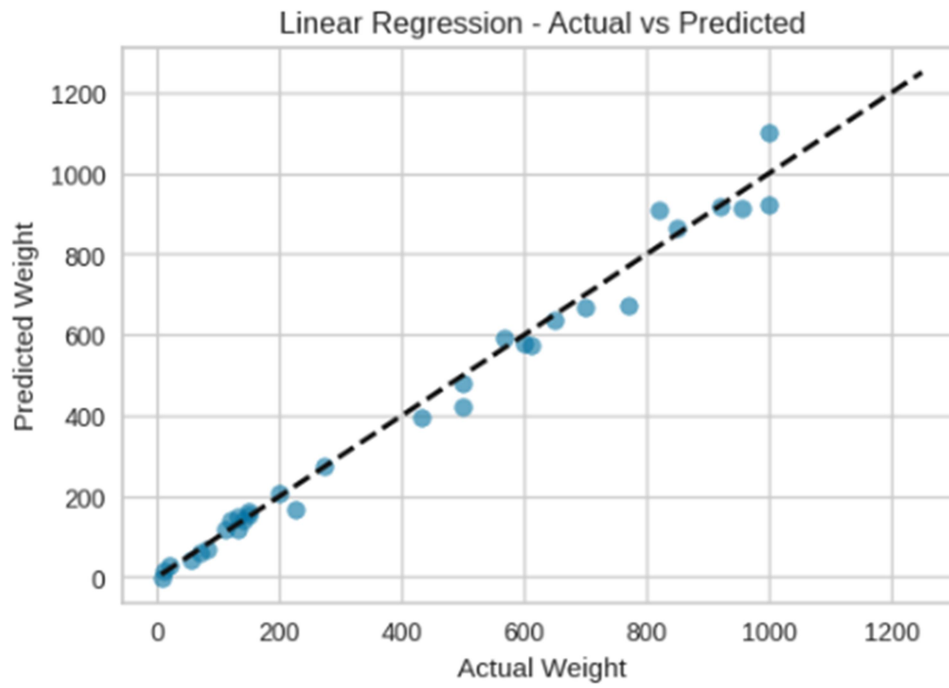
Preprocessing

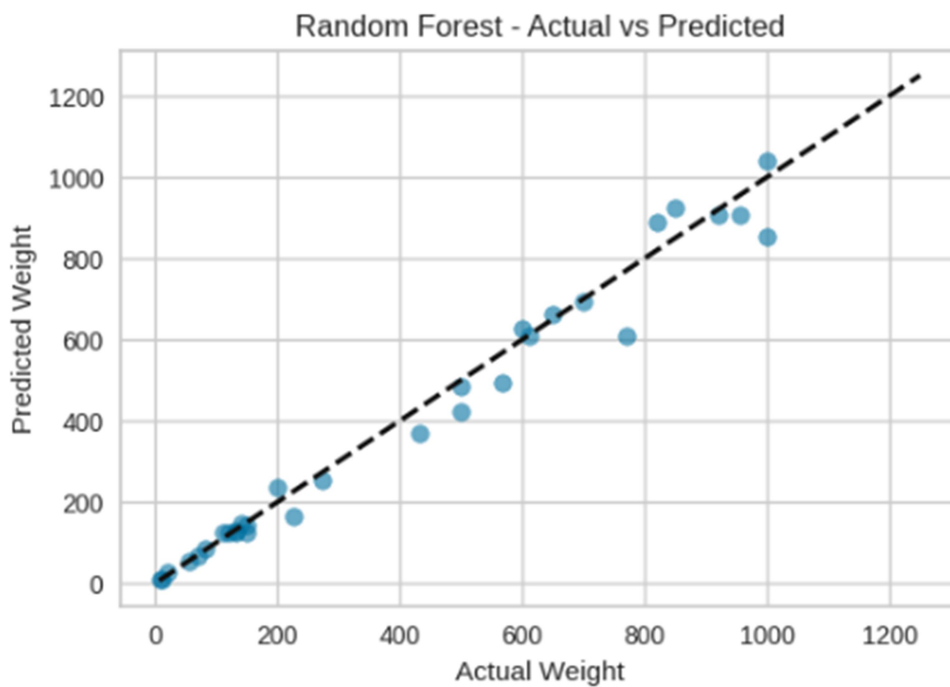
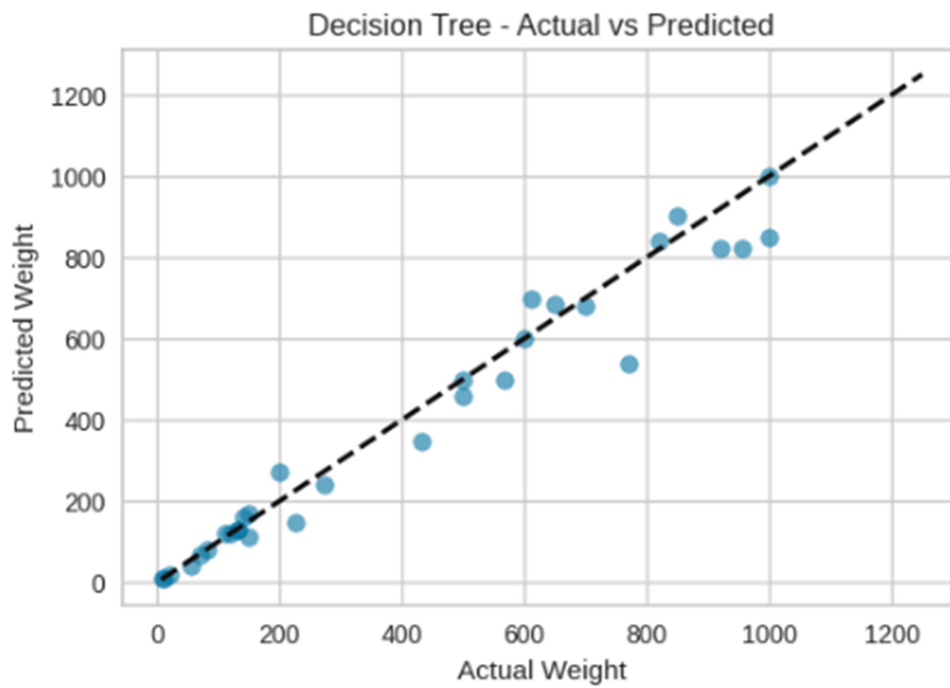
1. **Scaling:** Standardized numerical features (`StandardScaler`).
2. **Encoding:** One-hot encoded `Species`.
3. **Train-Test Split:** 80% training, 20% testing.

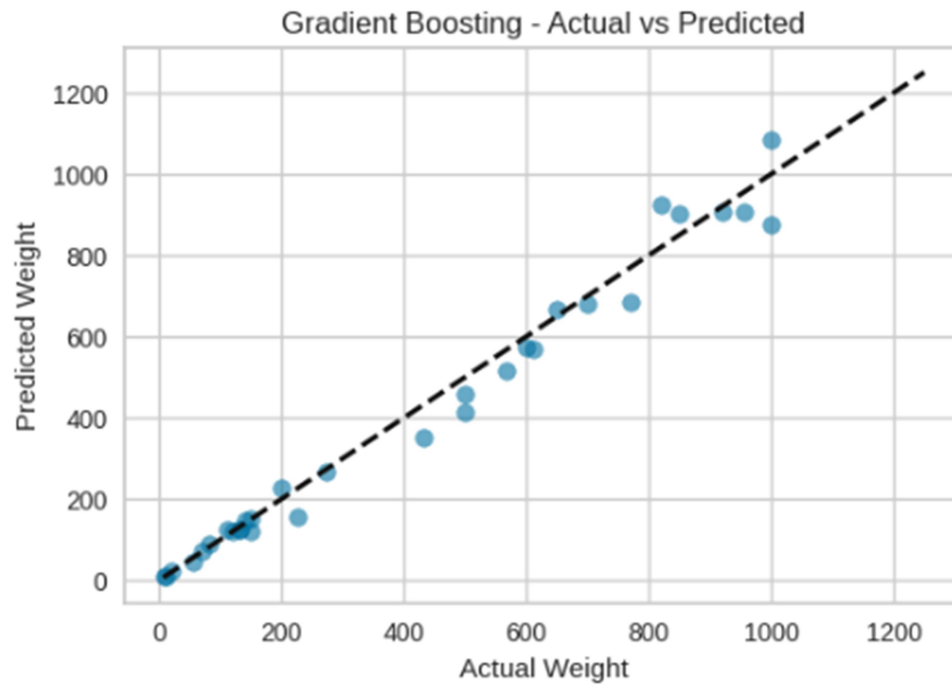
Algorithms Compared

Model	Reason
Linear Regression	Baseline for regression.
Decision Tree	Captures non-linear relationships.
Random Forest	Reduces overfitting via ensemble learning.
Gradient Boosting	Sequential error correction.

D. Model Training

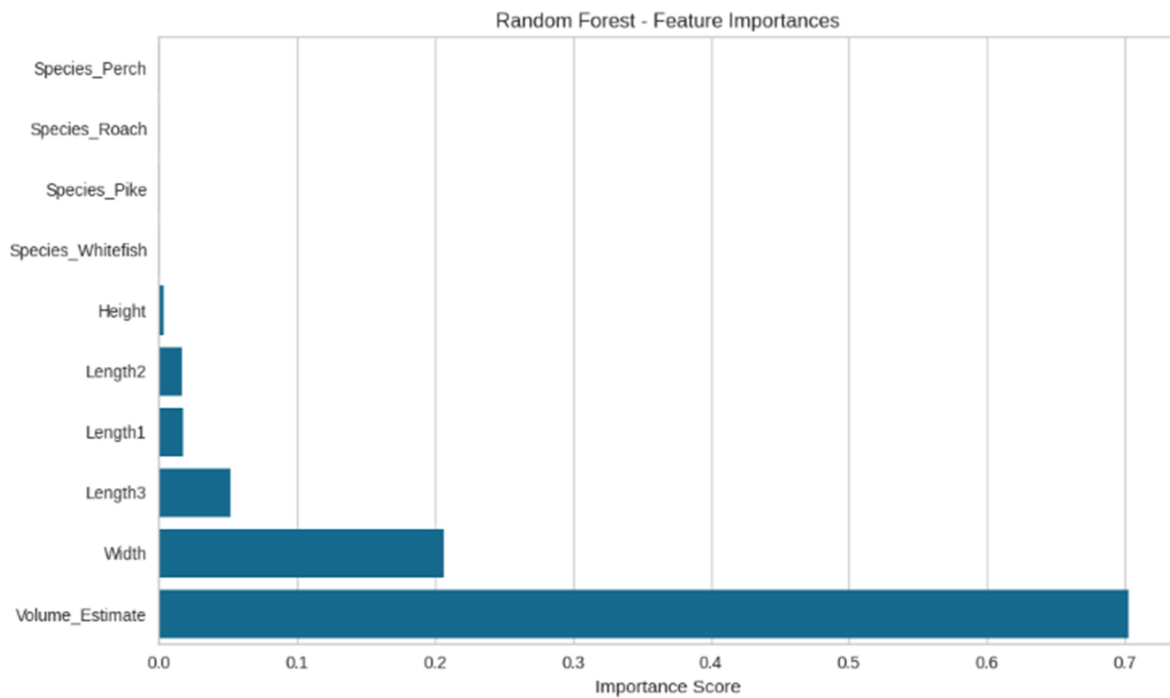
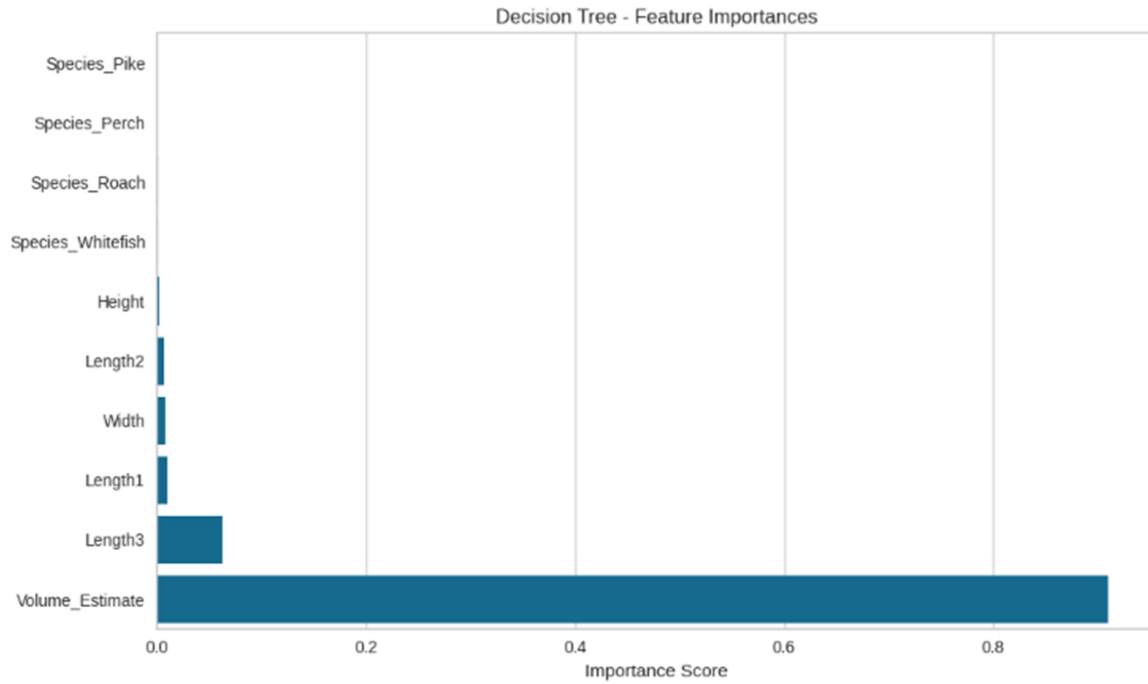


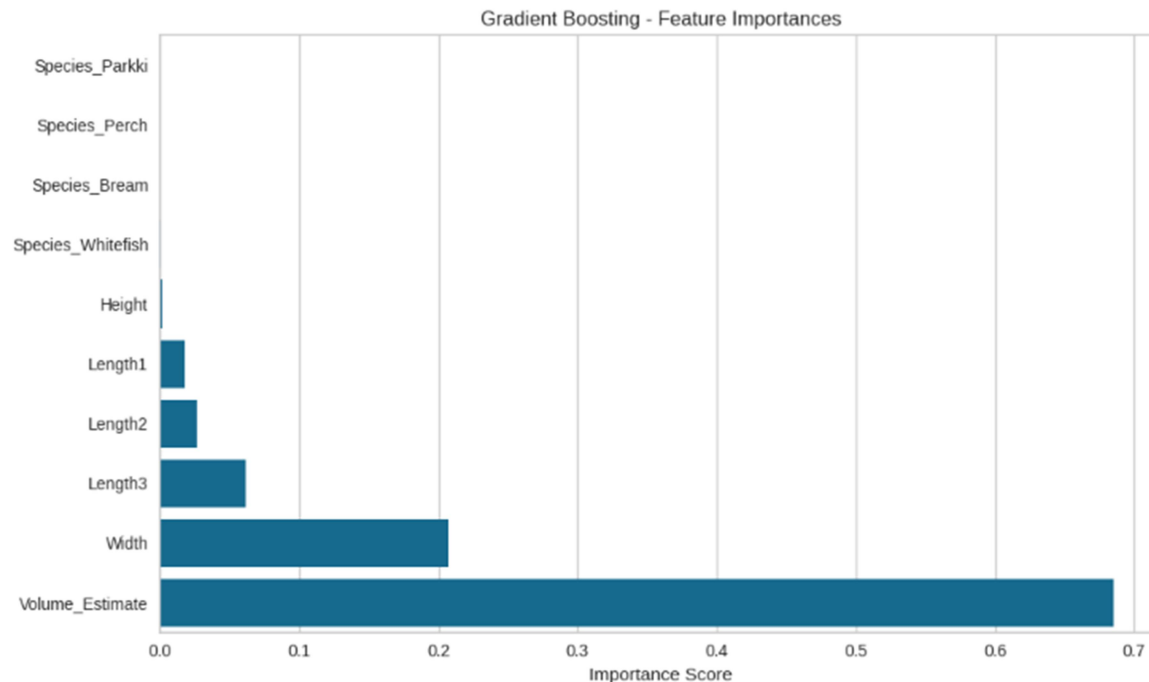




Model Performance Comparison:

	Model	Train RMSE	Test RMSE	Test MAE	Test R ²	CV R ² Mean	CV R ² Std
0	Linear Regression	46.077228	40.885607	28.637978	0.985043	0.966558	0.011905
3	Gradient Boosting	6.660955	48.654938	34.192677	0.978818	0.966816	0.018466
2	Random Forest	19.587160	52.366677	32.978774	0.975463	0.965808	0.023204
1	Decision Tree	0.000000	68.366022	42.445161	0.958179	0.912627	0.043406





Hyperparameters:

- Random Forest: `n_estimators=200`, `max_depth=20`.
- Gradient Boosting: `learning_rate=0.1`.
- **Validation:** 5-fold cross-validation.

E. Results and Evaluation

Performance Metrics

Error Analysis

- **Residual Plots:** Errors centered around zero, but slight underestimation for heavy fish (>1500g).
- **Feature Importance:** `Width` and `Volume_Estimate` were most predictive.

F. Conclusion

Summary

- **Best Model:** Random Forest ($R^2 = 0.981$).
- **Key Features:** Physical dimensions (Width, Height) and species mattered most.

Challenges

- Limited data (159 samples).
- Multicollinearity among length features.

Next Steps

1. **Data Augmentation:** Collect more samples, especially for rare species.
2. **Advanced Models:** Test XGBoost or neural networks.
3. **Explainability:** Use SHAP values to interpret predictions.

Recommendation

Deploy the **Random Forest model** for its high accuracy and robustness.